

10.HASH FUNCTION

```
#include <stdio.h>
#define MAX 20
int hashTable[MAX];
int m;
void insert(int key){
    int index = key % m;
    if(hashTable[index] == -1){
        hashTable[index] = key;
    }else{
        int i = 1;
        while(hashTable[(index + i) % m] != -1){
            i++;
        }
        hashTable[(index + i) % m] = key;
    }
}
void display(){
    printf("\nHash Table:\n");
    for(int i = 0; i < m; i++){
        if(hashTable[i] != -1)
            printf("Address %d : %d\n", i, hashTable[i]);
        else
            printf("Address %d : Empty\n", i);
    }
}
int main(){
    int n, key;
    printf("Enter size of hash table (m): ");
    scanf("%d", &m);
    printf("Enter number of employee records: ");
    scanf("%d", &n);
    for(int i = 0; i < m; i++)
        hashTable[i] = -1;
    printf("Enter %d employee keys (4-digit):\n", n);
    for(int i = 0; i < n; i++){
        scanf("%d", &key);
        insert(key);
    }
    display();
    return 0;
}
```

OUTPUT:

```
C:\Users\BMSCE\Desktop\1BF X + v
Enter size of hash table (m): 10
Enter number of employee records: 6
Enter 6 employee keys (4-digit):
1230
1247
1357
1789
1999
1555

Hash Table:
Address 0 : 1230
Address 1 : 1999
Address 2 : Empty
Address 3 : Empty
Address 4 : Empty
Address 5 : 1555
Address 6 : Empty
Address 7 : 1247
Address 8 : 1357
Address 9 : 1789

Process returned 0 (0x0)   execution time : 24.985 s
Press any key to continue.
|
```